

Evaluation of the Vpp and its uncertainty using an oscilloscope

Students of the academic year 2024/2025

09/06/2025



UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

**MEASUREMENTS FOR TELECOMMUNICATION
SYSTEMS** by Prof. Francesco Lamonaca
A.A. 2024/2025

This work is part of the project:

EXTENDABLE

NETWORK OF **EXTENDED REALITY-ENABLED**
LABORATORIES FOR REMOTE PRACTICAL TRAINING

Web-site: <https://extendable.dieti.unina.it/>



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Italiadomani
PIANO NAZIONALE
DI RIPRESA E RESILIENZA

**The Research Team involving Units from:
University of Calabria, Naples Federico II, and Sannio
is proposing the creation of a network of laboratories**



Process

Goals

Instruments

Questions?

What are the instruments?

How to link them?



Procedure

Sinusoidal Signal

**Oscilloscope
Configuration**

**Reading &
Evaluation**

**Propagation
of Uncertainty**

Conclusion

Hard and soft skills



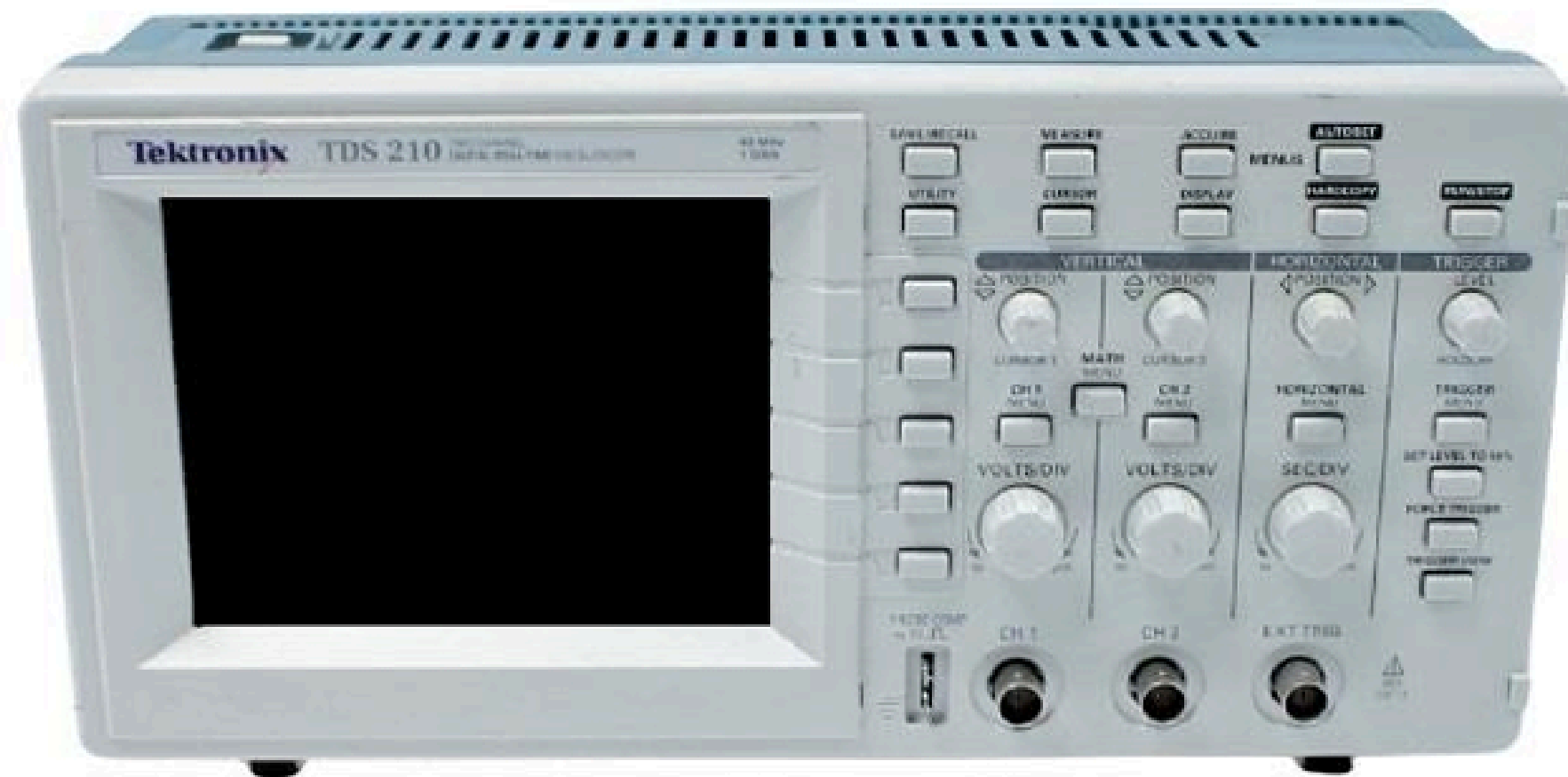
Goals

Evaluate the peak-to-peak voltage (V_{pp}) of a sinusoidal waveform using the oscilloscope display

Evaluate the uncertainty of the measurement using basic analysis and statistical assumptions.

What are the instruments?

Oscilloscope: Tektronix TDS 210



Bandwidth:

- 60 MHz (TDS 210)
- Selectable 20 MHz bandwidth limit
- Sample Rate: 1 GS/s
- Record Length: 2,500 points for each channel



What are the instruments?

Function Generator: Keysight 33220A



Key Features:

- 20 MHz Sine and Square waveforms
- Pulse, Ramp, Triangle, Noise, and DC waveforms
- 10 mVpp to 10 Vpp amplitude range
- AM, FM, PM, FSK, and PWM modulation types
- 14-bit, 50 MSa/s, 64 k-point arbitrary waveforms



How to link them?

One BNC-BNC coaxial cable



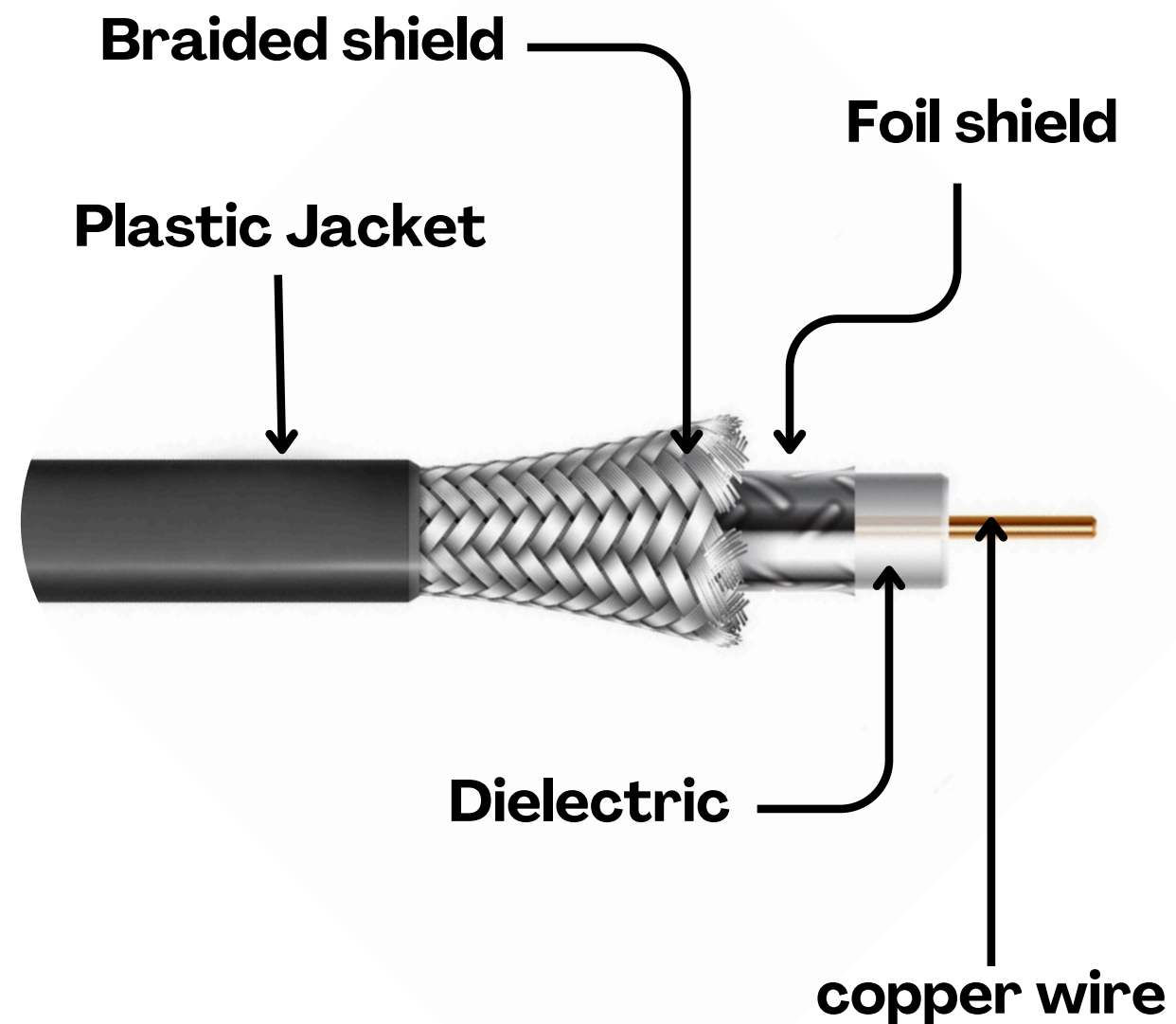
Why in parallel?

- Instruments share the same Voltage

Why a Coaxial Cable?

- Impedance closely matches the output impedance of the connected instruments
- Shield from external noise

How to link them?



Why the Plastic jacket?

- Shield from external interferences

Why the Braided shield?

- Prevent that the cable act as an antenna
- Protecting the signal from external noise
- Acting as a reference ground for the instruments.

Procedure

Set of the Function Generator

Sinusoidal Signal

f

2 kHz

0 V

Offset

A

2 V_{pp}

**High
Impedance**

Out-Imp

Procedure

Oscilloscope Configuration

Display settings:

Trigger

V/div

500 mV

0 V

Level

s/div

100 μ s

CH1

Channel

Vertical
Position

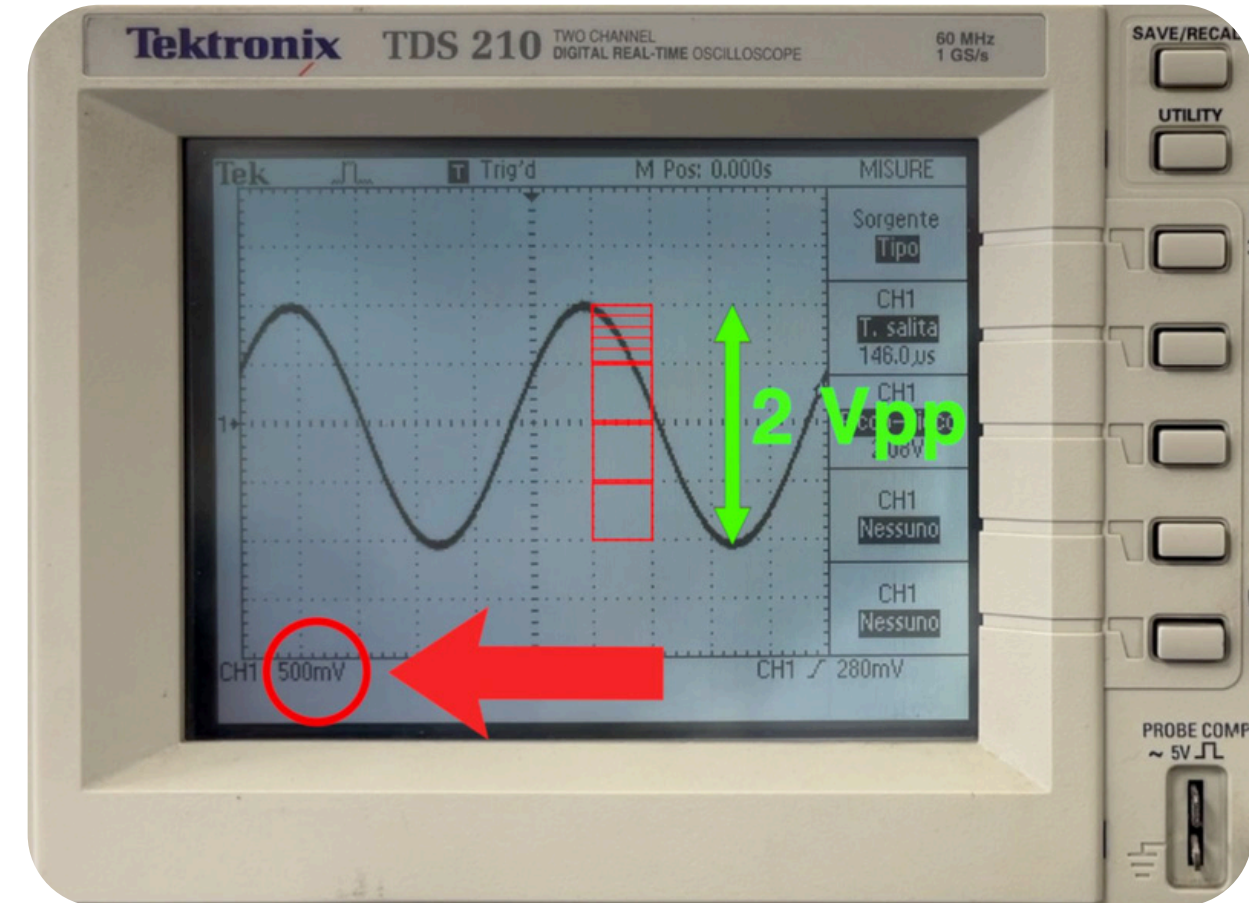
0 V

50% of
memory

Position

Reading & Evaluation

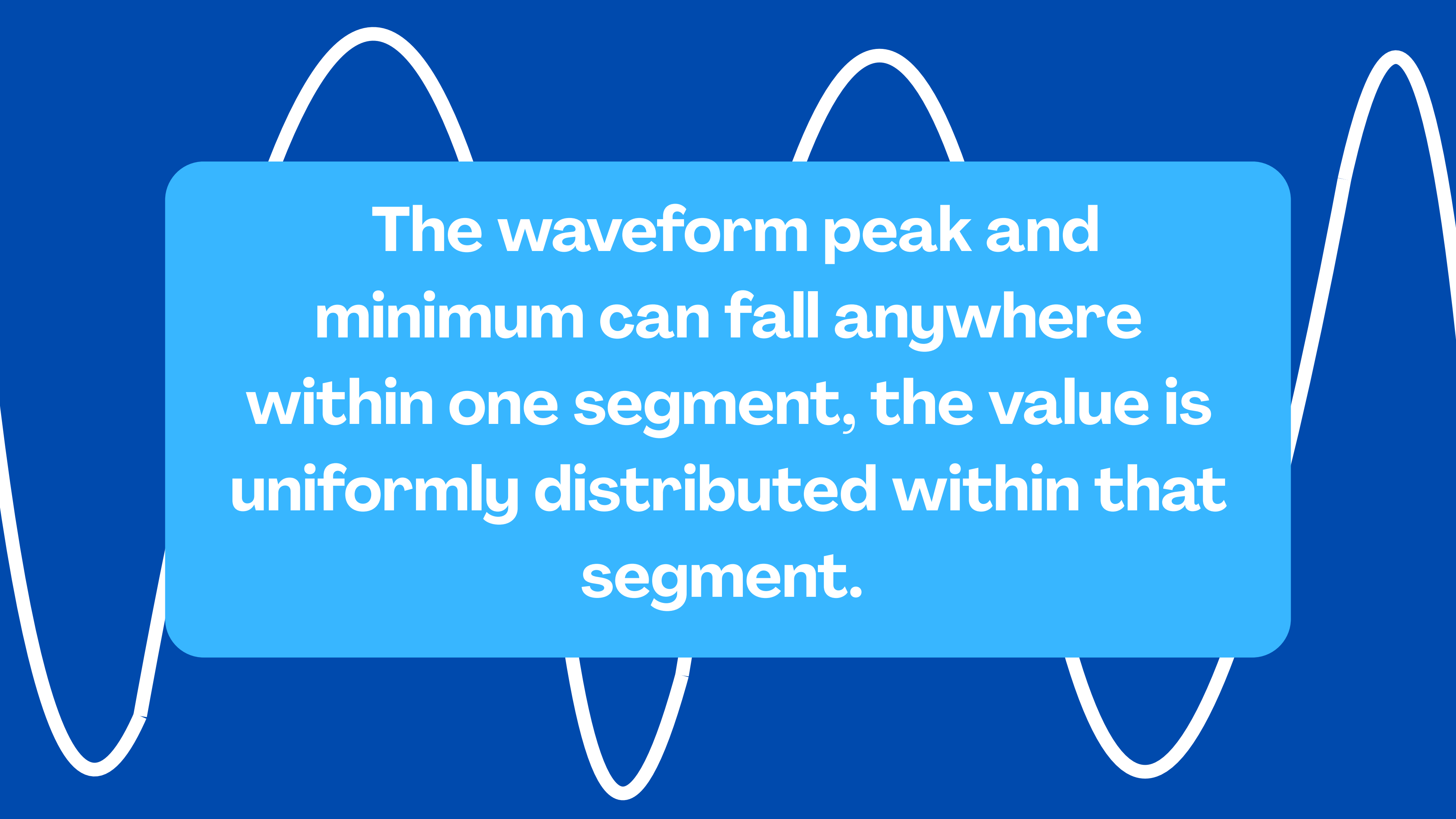
Reading Peak-to-Peak
Voltage (V_{pp})



Each square = 500 mV
 $V_{pp} = 4 \times 500 \text{ mV} = 2 \text{ V}$

A white sine wave oscillates across a solid blue background. In the center, a light blue rounded rectangle contains white text. The wave passes behind the rectangle, with its peaks and troughs visible above and below the box.

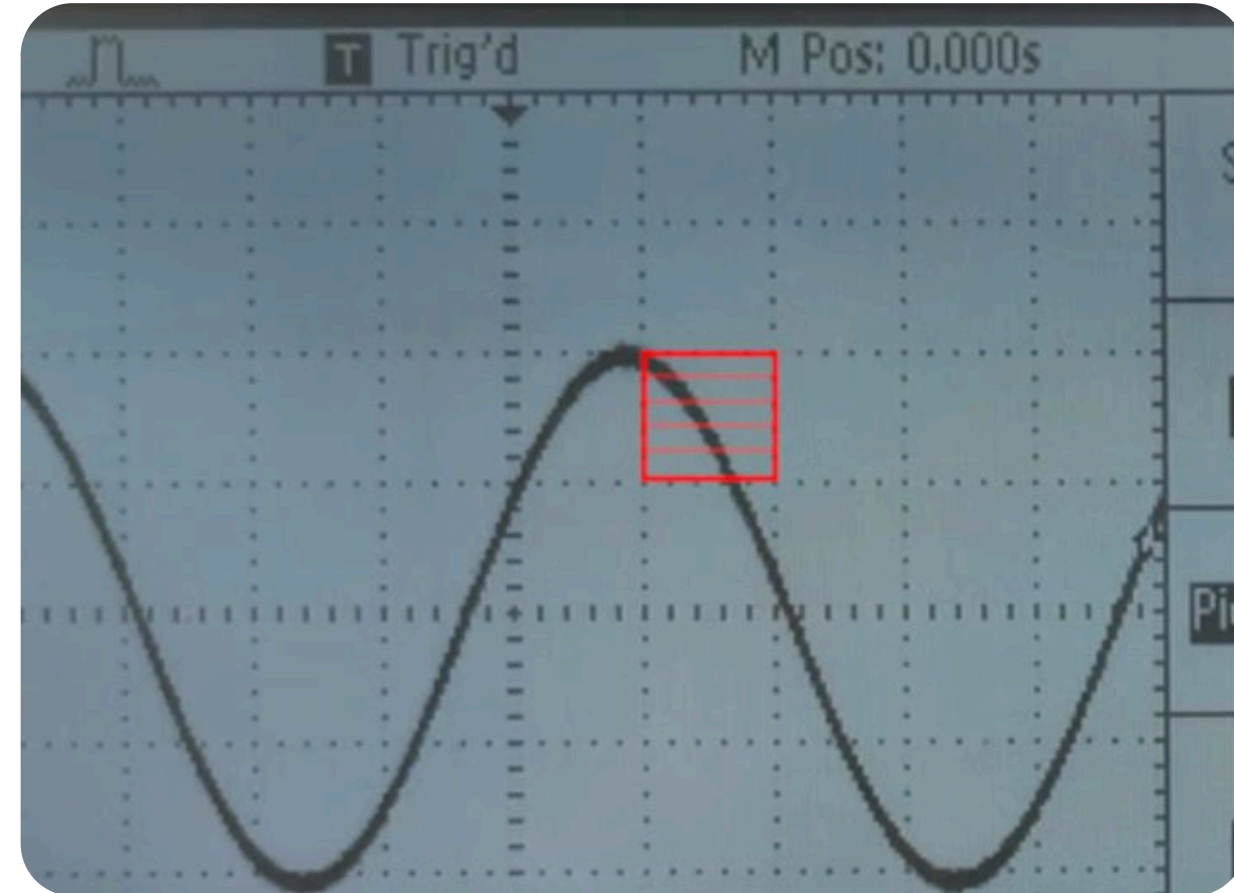
**Let's first evaluate
the uncertainty of Peak and
Minimum Values**



The waveform peak and minimum can fall anywhere within one segment, the value is uniformly distributed within that segment.

Reading & Evaluation

Uncertainty of Peak and Minimum Values



Each square is divided into 5 segments . One is equals to $500 \text{ mV} / 5 = 100 \text{ mV}$

Reading & Evaluation

Uncertainty of Peak and
Minimum Values

Type B

$$u = \frac{(\text{segment} / 2)}{\sqrt{3}}$$
$$= \frac{\frac{100}{2}}{\sqrt{3}} \approx 28.87 \text{ mV}$$

$$u_{Vmax} = u_{Vmin} = u$$

Reading & Evaluation

Uncertainty of V_{pp}

Propagation of uncertainty formula

$$u_f = \sqrt{\left(\frac{\partial f}{\partial x_1} u_{x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2} u_{x_2}\right)^2 + \dots + \left(\frac{\partial f}{\partial x_n} u_{x_n}\right)^2}$$

In our case :

- $V_{pp} = V_{max} - V_{min}$
- $u_{V_{max}} = u_{V_{min}} = u$

Reading & Evaluation

Uncertainty of V_{pp}

Propagation of
uncertainty formula

$$u_{V_{pp}} = \sqrt{u^2 + u^2} = \sqrt{2(u)^2} =$$

$$= \sqrt{2 \left(\frac{50}{\sqrt{3}} \right)^2} \approx 40.82 \text{ mV}$$

Conclusion

Hard Skills Acquired

- **Instruments Set up**
- **Vpp Evaluation using an Oscilloscope**
- **Uncertainty evaluation**

Conclusion

Hard Skills Acquired

- **Data Analysis and Interpretation**
- **Oscilloscope and Function Generator usage**

Conclusion

Soft Skills Developed

- **Experiment Organization**
- **Teamwork and Collaboration**
- **Communication**

Conclusion

Soft Skills Developed

- **Problem-Solving**
- **Critical Thinking**
- **Time Management**
- **Attention to details**

Thank you for your attention

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